Faithful point shadows and automorphisms of twenty-four Higman–Sims geometries

L091 finite classification component

Let $G = \mathrm{HS}$, of order 44 352 000, in the acquired faithful permutation action on $\Omega = \{1, \ldots, 100\}$. Let N be the acquired permutation group inducing the complete $\mathrm{Aut}(G)$, of order 88 704 000, with $C_N(G) = 1$. We use the fixed representatives in `full_fusion03/result.g`, their already established geometry and flag-transitivity properties, and their actual parabolic subgroups $P_1, \ldots, P_r$. Identifiers below refer to that file, not to historical diagram numbers. Automorphisms preserve every type; correlations may permute types. Put $B = \bigcap_i P_i$ and $A = \mathrm{Aut}(G)$.

Theorem 1. *Under the acquired finite-group inputs and the exact graph-automorphism premise stated below, the full groups of the following geometries are:*

Representatives	$\mathrm{Aut}(\Gamma)$	$\mathrm{Cor}(\Gamma)$
$19, 20, 41, 44, 47, 79, 80$	G	G
67	G	A
$83, 84$	G	$G \times C_2$
$6, 30, 31, 40, 75, 76$	A	A
$28, 60, 72, 77, 81, 82, 85, 86$	A	$A \times C_2$

These are all twenty-four representatives having a type of size 100*. The assertion is about full groups, not only normalizers of the specified G action.*

Restriction to point shadows

For each type a of size 100, the actual subgroup P_a fixes exactly one point $u_a \in \Omega$. Its order equals $|G|/100$, so it is exactly G_{u_a}. Thus $P_a g \mapsto u_a^g$ is a bijection from that type to Ω. For the base object of type j, its shadow on type a is $u_a^{P_j}$. This follows directly from right-coset incidence: the incident objects of type a are $P_a p$ for $p \in P_j$.

Lemma 2. *Let a transitive type have base stabilizer P. An equivariant encoding of its objects is injective if and only if the stabilizer in G of its base encoding is P. Once certain types are pointwise fixed by an automorphism, so are all objects encoded injectively by their shadows or by their sets of neighbours in those types.*

Proof. An equivariant encoding is the map from $P\backslash G$ to the orbit of its base value. Its fibres have size $[G_{\mathrm{value}} : P]$. An automorphism fixing the encoded types fixes every such set or tuple of their objects, and therefore every object having a unique encoding. This also proves the iterative assertion. □

The literal calculation checks that every 100-object type is determined by its shadow on any other such type, including itself. For all other types start with the ordered tuple $(u_a^{P_j})_a$. Its stabilizer is the intersection of the set stabilizers. Fifteen families already have every object determined this way. In the others, adjoin sets of encoded neighbours from already identified types. If type k is identified, the neighbours of the base type-j object are precisely the orbit

of its base type-k encoding under P_j. The program checks the orbit length $[P_j : P_j \cap P_k]$, then intersects the current encoding stabilizer with its set stabilizer. Only a type whose resulting stabilizer equals its actual P_j is admitted for use in later steps. There are 98 types in total. All are identified. Representative 40 needs two successive neighbour steps; no unproved direct-shadow injectivity is used there. Lemma 2 proves faithful restriction of the full typed automorphism group to each of the 38 possible 100-object anchors.

The intrinsic graph bound

The orbits of a point stabilizer on Ω have sizes $1, 22, 77$. Transport its orbit of size 22 to obtain a simple undirected graph Δ of degree 22. The two unordered pair orbits have sizes 1100 and 3850. These assertions are checked in the actual action. For a type-j shadow $S = u_a^{P_j}$ let e be the number of edges of Δ inside S, and put $f = \binom{|S|}{2} - e$. Double counting pairs and incident objects gives the two common-incidence counts

$$\frac{[G : P_j]e}{1100}, \qquad \frac{[G : P_j]f}{3850}. \tag{1}$$

This formula counts objects, so repeated point shadows cause no error. For every anchor at least one type has unequal counts in (1). Consequently the vector of common-incidence counts over all types partitions the distinct point pairs into exactly these two orbits. An arbitrary correlation permutes the coordinates of this vector and maps an anchor to another 100-object type. The two pair relations have different valencies, so the degree-22 relation is preserved. Thus every correlation induces an isomorphism of the corresponding copies of Δ.

The complete nauty computation for this single 100-vertex graph gives order 88 704 000. Every returned generator and every acquired generator of N is checked on all adjacencies. Hence the full graph group is exactly N. The literal permutation calculation also checks $N' = G$; in particular G is characteristic in the graph group. For any correlation c and any $g \in G$, its induced point action therefore makes cgc^{-1} equal on the target anchor to some $h \in G$. The typed automorphism $h^{-1}cgc^{-1}$ is trivial on that anchor. Faithfulness from Lemma 2 makes it trivial everywhere. We have proved that every full correlation normalizes the specified G.

Exact lifts and the group structures

Write S and T for the ordered and setwise stabilizers, respectively, of $(P_1, \dots, P_r)$ in the complete group $N = \mathrm{Aut}(G)$. An element $t \in T$ with $P_i^t = P_{\pi(i)}$ acts by

$$P_i g \longmapsto P_{\pi(i)} g^t. \tag{2}$$

This is well defined and preserves incidence in both directions. It fixes the base chamber setwise. Conversely, a correlation normalizing G can first be moved by G so that it fixes that chamber. Its induced automorphism of G then belongs to T; faithfulness of the action and the coset description give (2). The ordered analogue gives S. Consequently

$$|\mathrm{Aut}(\Gamma)| = |G : B||S|, \qquad |\mathrm{Cor}(\Gamma)| = |G : B||T|. \tag{3}$$

The complete type-lift calculation tests every type permutation, using subgroup orders and orders of pairwise intersections only to reject impossible cases. For each remaining permutation it successively transports actual subgroups in N, replacing the transporter group by the stabilizer of each target already placed. The transporter-coset induction proves that this loses no solution. Every positive witness is checked on the whole ordered family. The group generated by S and these witnesses has order $|S|$ times the number of successful type permutations, and stabilizes the family. It is therefore exactly T.

For completeness, let $C = C_{\mathrm{Cor}(\Gamma)}(G)$. Centralizing correlations are the maps obtained from inner elements $t \in T \cap G$ in (2), followed by the inverse right action of t. Their kernel is B, and all centralizing correlations arise this way by moving the image chamber back. Thus $C \cong (T \cap G)/B$ (an opposite-group convention gives the same isomorphism). Direct calculations give $S \cap G = B$, so C meets the typed automorphism group trivially. The quotient above is always trivial or of order two; its actual quotient generators are retained. The conjugation image of $\mathrm{Cor}(\Gamma)$ in $\mathrm{Aut}(G)$ has order $|\mathrm{Cor}(\Gamma)|/|C|$, hence is G or A.

If $C = 1$, this identifies the full group with that image. If $|C| = 2$, it is central in the full group, and in every such row the typed automorphism group has order $|\mathrm{Cor}(\Gamma)|/2$. Their trivial intersection and full product order give the direct product asserted in Theorem 1. In particular the factor C_2 is not inferred merely from a doubled cardinality. The exceptional row 67 has typed group G, trivial C, and full conjugation image A.

Inputs, attribution and remaining scope. The acquired complete HS action, family enumeration, and its exact Aut/type quotient are explicit premises. GAP's finite permutation and subgroup algorithms and nauty's complete graph-automorphism algorithm are retained computational inputs. The scripts retain the literal shadow encodings, positive type lifts and every rejected type permutation; no subgroup or geometry census is repeated. Small literal incidence controls test restriction, iterative shadows, type interchanges and a nonfaithful duplicate-shadow example. The shadow and transporter arguments are elementary and overlap the accepted common incidence theory. The historical L096 naming treatment retained the source's full typed-group assertion; it is not an upper-bound premise here. This component supplies a new check of the required bounds, not a novelty claim for the original results.

Representatives $14, 15, 35, 36$ have separate acquired full-group proofs. The remaining 58 representatives are outside the present result. All 86 source-diagram identifications and the global source synthesis remain separate. The source is D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236, together with the explicitly retained 6 October 2003 supplementary diagrams. No whole-source, measurement, formalization or new-question credit is claimed by this component.